

Quiz 3: Non-local Games

Recall that the CHSH game is the non-local game where Alice and Bob are given uniformly random questions $x, y \in \{0, 1\}$, respectively, and they produce answers $a, b \in \{0, 1\}$. They win if $x \cdot y = a + b \pmod 2$.

Question 1. Which of the following statements is true?

- (a) Randomized strategies are better than deterministic strategies.
- (b) Quantum strategies are better than randomized strategies.

Question 2. Suppose that Alice and Bob share a 2-party state ρ_{AB} . Upon receiving questions, they perform measurements μ_A^x and μ_B^y , respectively. Which of the following statements are true?

- (a) The order in which Alice and Bob perform their measurements does not affect their probabilities.
- (b) The outcomes of Alice's and Bob's measurements are independent if ρ_{AB} is not entangled.

Question 3. Suppose that Alice and Bob share a quantum state of the form

$$\rho_{AB} = \sum_{\lambda} p_{\lambda} \rho_{A,\lambda} \otimes \rho_{B,\lambda},$$

where $\rho_{A,\lambda}$ and $\rho_{B,\lambda}$ are quantum states for all λ and $\{p_{\lambda}\}_{\lambda} \in \Lambda$ is a probability distribution. What is the optimal winning probability?

- (a) $\frac{1}{2}$
- (b) $\frac{3}{4}$
- (c) $\frac{1}{2} + \frac{1}{2\sqrt{2}}$
- (d) 1

Question 4. Suppose that Alice and Bob share the pure quantum state

$$|\psi_{AB}\rangle = \frac{1}{2}|0\rangle \otimes (|0\rangle + |1\rangle) + \frac{1}{2}|1\rangle \otimes (|0\rangle - |1\rangle).$$

What is the optimal winning probability?

- (a) $\frac{1}{2}$
- (b) $\frac{3}{4}$
- (c) $\frac{1}{2} + \frac{1}{2\sqrt{2}}$
- (d) 1